

Tank Arena — Map Analysis

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1. Map Dimensions

WORLD WIDTH

1600 px

Horizontal axis

WORLD HEIGHT

1200 px

Vertical axis

DEFAULT OBSTACLE SIZE

64 × 64 px

Width & height default

COORDINATE SYSTEM

Center origin

x,y = center of obstacle

All obstacle `x` and `y` values are **center positions**. The renderer calculates the top-left corner as `x - width/2`, `y - height/2`.

2. Obstacle Types Reference

Type	HP	Shots to destroy (default bullet: 20 dmg)	Blocks Players	Blocks Bullets	Default assetId	Behaviour
bush	34	2	No	Damages it	bush_01_rounded_dense	Soft cover Players pass through. Bullets deal

Type	HP	Shots to destroy (default bullet: 20 dmg)	Blocks Players	Blocks Bullets	Default assetId	Behaviour
						damage and eventually destroy it.
decoration	9999	∞	No	No	decoration_01_spiky_organic	Visual only Purely cosmetic. Does not block anything.
wood	68	4	Yes	Yes	wood_barricade	Destructible Solid cover. Can be shot down after sustained fire.
rock	102	6	Yes	Yes	rock_block	Destructible Harder solid cover. Takes more fire to break.
steel	9999	∞	Yes	Yes	steel_block_01	Indestructible Permanent hard anchor. Cannot be destroyed.
mirror	9999	∞	Yes	Reflects	mirror_panel_01	Reflector Indestructible. Reflects bullets off the hit face.

BUSH VS DECORATION — KEY DIFFERENCE

- **Bush** — has collision with bullets (HP depletes), can be destroyed. Used as soft cover that eventually disappears.
- **Decoration** — completely ignored by all game logic. Pure visual dressing.

MIRROR GEOMETRY TIP

Mirrors use non-square dimensions to define the reflective face direction:

- `width > height` (e.g. 140×18) → horizontal mirror, reflects bullets traveling vertically.
- `height > width` (e.g. 18×130) → vertical mirror, reflects bullets traveling horizontally.

3. Visual Asset IDs

BUSH VARIANTS (4)

assetId	Description
bush_01_rounded_dense	Dense round shrub (default)
bush_02_irregular_leafy	Leafy irregular shape

SOLID OBSTACLES

Type	assetId
wood	wood_barricade
rock	rock_block

assetId	Description	Type	assetId
bush_03_compact_arcade	Compact arcade-style	steel	steel_block_01
bush_04_wide_low	Wide, low-profile	mirror	mirror_panel_01

DECORATION VARIANTS (14)

assetId	assetId
decoration_01_spiky_organic	decoration_08_tall_grass_wildflowers
decoration_02_two_lobed	decoration_09_cactus_flowers
decoration_03_pink_yellow_flowers	decoration_10_reed_patch_orange
decoration_04_grass_blue_flowers	decoration_11_fern_star
decoration_05_wild_red_flowers	decoration_12_clover_patch
decoration_06_sharp_grass_pink_yellow	decoration_13_dry_grass_mix
decoration_07_leafy_blue_flower	decoration_14_vine_swirl

4. Current Map Layout (Overhead Overview)

SVG schematic — not to pixel scale. Numbers next to elements show approximate coordinates. Map is 1600×1200 px.



5. Player Spawn Points

Players are assigned spawns in round-robin order (`players.size % 15`). Assigned as players join, not re-assigned on respawn.

#	X	Y	Zone
1	150	150	Top-left corner
2	1450	150	Top-right corner
3	150	1050	Bottom-left corner
4	1450	1050	Bottom-right corner

#	X	Y	Zone
5	800	120	Top center edge
6	800	1080	Bottom center edge
7	120	600	Left center edge
8	1480	600	Right center edge
9	400	300	Upper-left quad
10	1200	300	Upper-right quad
11	400	900	Lower-left quad
12	1200	900	Lower-right quad
13	550	550	Center-left
14	1050	550	Center-right
15	800	400	Center-top

6. Power-up Spawn Locations

Type	X	Y	Radius	assetId	Effect
triple_shot	800	520	18 px	power_triple_shot	3 bullets per shot at spread angles
shotgun	730	650	18 px	power_shotgun	5 pellets, shorter range
grenade	870	650	18 px	power_grenade	Slow projectile, AOE explosion radius 100
laser	800	720	18 px	power_laser	2 shots, 1000 ms beam, 1200 max distance

All 4 power-ups are clustered in the lower-center area (730–870, 520–720). Players must converge to the center to collect them.

7. JSON Structure Reference

The complete `GameMap` JSON shape sent to clients via `gameJoined` :

```
{
  "width": 1600,
  "height": 1200,
  "obstacles": [
    {
      "id": "uuid-v4",
      "type": "rock" | "wood" | "steel" | "bush" | "decoration" | "mirror",
      "assetId": "rock_block",          // optional, falls back to type default
      "x": 360,                        // CENTER x position
      "y": 430,                        // CENTER y position
      "width": 64,                     // obstacle width (default 64)
      "height": 64,                    // obstacle height (default 64)
      "hp": 102,                       // current HP
      "maxHp": 102,                    // max HP
      "healthRatio": 1.0,               // hp / maxHp (0.0 - 1.0)
      "destructible": true
    }
  ],
  "powerUps": [
    {
      "id": "uuid-v4",
      "type": "triple_shot" | "shotgun" | "grenade" | "laser",
      "assetId": "power_triple_shot",
      "x": 800,
      "y": 520,
      "radius": 18                      // pickup radius in pixels
    }
  ]
}
```

OBSTACLE HP & DESTRUCTIBLE REFERENCE (FOR JSON AUTHORS)

type	hp / maxHp	destructible	healthRatio
bush	34	true	1.0
decoration	9999	false	1.0
wood	68	true	1.0
rock	102	true	1.0
steel	9999	false	1.0

type	hp / maxHp	destructible	healthRatio
mirror	9999	false	1.0

MIRROR ORIENTATION RULES

Desired orientation	width	height	Reflects
Horizontal panel (—)	e.g. 140	18	Bullets moving up/down
Vertical panel ()	18	e.g. 130	Bullets moving left/right
Diagonal-ish	~equal	~equal	Depends on hit face

8. Prompt Template for Custom Scenarios

Paste this template into a new conversation and fill in the placeholders:

Create a Tank Arena map scenario called **[SCENARIO NAME]**.

Map size: 1600 × 1200 px. Obstacle positions are center coordinates.

Theme: **[e.g. "abandoned factory", "jungle ruins", "urban street grid", "symmetrical sniper duel"]**

Layout intent:

- Number of lanes: **[e.g. 3 horizontal + 2 vertical]**
- Center area: **[open / contested / blocked]**
- Preferred obstacle density: **[sparse / medium / dense]**
- Symmetry: **[yes (left-right) / yes (4-way) / no]**

Obstacle types to include:

- bush: **[count or "many" / "few" / "none"]**
- wood: **[count]**
- rock: **[count]**
- steel: **[count, these are permanent anchors]**
- mirror: **[count, specify horizontal/vertical/mixed]**
- decoration: **[count, visual only, no collision]**

Special constraints:

- Keep the 4 corner spawn zones (within ~200 px of each corner) free of solid obstacles.
- Keep at least one clear path through the horizontal center lane (y ≈ 500–700).
- Power-up area: **[keep center / spread across map / remove and relocate]**

Output a JSON object with this exact structure:
{

```
"obstacles": [ { "type", "x", "y", "width"?, "height"?, "assetId"? } ... ],
"powerUps": [ { "type", "x", "y" } ... ]
}
```

Do not include "id", "hp", "maxHp", "healthRatio", or "destructible" — the server fills those from type defaults.

QUICK SCENARIO EXAMPLES YOU CAN USE DIRECTLY

Scenario name	Key characteristics
Urban Grid	3 vertical street corridors with wood barricades and steel anchors. Rock intersections at crossings. Bushes in alleys.
Sniper Duel	Long horizontal mirrors creating ricochet chambers. Very few solid obstacles. Power-ups at extreme edges.
Jungle Fortress	Dense bush perimeter, rock cluster center, steel bunker in middle. Only 4 spawn corners open.
Symmetrical Clash	Perfect 4-way symmetry. Identical cover lines per quadrant. Central steel tower unreachable without flanking.
Open Field	Minimal obstacles. All 4 power-ups spread to corners. Long-range and dash gameplay dominates.